TASK 5

Modelling Problem Specification

Smart Energy Forecasting for the CeDInt Building at UPM

1. Purpose

This document defines the first forecasting cases to be investigated by the modelling team. It translates the client-facing Product Backlog into a consistent technical problem definition before model training. The objective is to forecast energy-related behaviour, compare results fairly, and provide evidence that can support energy analysis and anomaly investigation.

2. Modelling principles and scope

- **Frozen source dataset:** Use the approved, versioned frozen dataset as the reference input. Never overwrite or manually modify it. Any hourly aggregation, filtering, feature engineering or target-specific preparation must create a separate, documented derived dataset.

- **Spatial scope:** Consider Storey 1 zones NE, NW, SE and SW. Include a zone/series only where the required target has sufficient valid coverage. Report exclusions and uneven coverage rather than silently filling gaps.

- **Forecast horizon:** The forecasts will have an hourly resolution with a 24-hour horizon to allow both the identification of variations during the day and the anticipation of the next day's workload.

- **Primary modelling resolution:** Use hourly targets for the first common comparison. The recent 2025 observations are minute-level and must be aggregated in a separate modelling dataset. Aggregation must respect each variable's semantics.

- **Temporal regimes:** The historical period (2020–2023) and recent period (2025) are separated by a long data gap. Do not create a continuous sequence across this gap or assume the two regimes are directly comparable without checking.

- **Forecasting, not operational control:** The work supports analysis and forecasts. It does not modify the live building pipeline, control equipment, or provide real-time streaming.

3. Three target variables

The team will investigate three distinct targets, with a clear priority order. energy_Wh is the main client-facing target. W and I are secondary electrical-behaviour targets that can provide complementary information for peak analysis and anomaly investigation. Their inclusion is conditional on data coverage, confirmed semantics/units, and sufficient modelling value.

Target 1 - energy_Wh

PRIMARY

Definition: Active energy consumed during the measurement interval.

Forecasting case: Forecast total active energy consumed during the next hourly interval (first then next 24h by hour), in Wh, for each eligible zone/series.

Rationale: This is the most direct match to the Product Backlog requirement to forecast energy consumption. It supports anticipated consumption and can provide an expected-consumption reference for investigating unusually high energy use.

Preparation rule: Aggregate interval energy by summing valid, non-overlapping intervals belonging to the target hour. Preserve energy_source provenance and document the treatment of period_start and unknown records. Do not assume all source categories have equal reliability.

Limitation / validation: Confirm interval boundaries and completeness before summing. Missing intervals must not be silently treated as zero. Evaluate whether counter, spread and power source categories behave comparably.

Target 2 - W

SECONDARY

Definition: Power-related measurement identified by the source column W; its unit and precise measurement semantics must be confirmed in the data documentation.

Forecasting case: Forecast the next-hour power behaviour using a defined hourly statistic (initial candidate: hourly mean), with the statistic documented and used consistently.

Rationale: Power complements energy: it can reveal demand peaks and short-lived high-load behaviour that may be less visible in an hourly energy total. Unusual observed-versus-expected power can support anomaly investigation.

Preparation rule: Do not sum instantaneous power readings. If W represents instantaneous power in watts, calculate a time-weighted or coverage-aware hourly mean where sampling is adequate; otherwise document the chosen alternative.

Limitation / validation: Confirm units, sampling interpretation and interval coverage first. A simple arithmetic mean may be misleading when intervals are irregular or data is missing.

Target 3 - I

EXPLORATORY SECONDARY

Definition: Current-related measurement identified by the source column I; its unit and exact interpretation must be confirmed.

Forecasting case: Forecast a documented hourly current statistic (initial candidate: hourly mean) for eligible electrical series, only if data is sufficiently complete and meaningful.

Rationale: Current provides a complementary view of electrical load. It may help characterise unusual operating patterns alongside energy and power, and tests whether a second electrical signal adds useful evidence for anomaly investigation.

Preparation rule: Use an hourly statistic appropriate to the measurement semantics and sampling. Keep the same time boundaries and eligibility rules used for the corresponding series wherever possible.

Limitation / validation: Current is not a substitute for energy consumption and may be redundant with power. Keep this target only if data quality and validation results justify the added effort.

4. Candidate input features

Candidate inputs will be tested only if available at prediction time, sufficiently reliable, and justified by the data:

- Lagged values of the same target and relevant historical energy/power measurements.
- Calendar features derived from the Madrid-local timestamp: hour, day of week, month, weekend and relevant holidays/calendar information where available.
- Zone and sensor/series identifiers, where appropriate and where enough observations exist to learn stable patterns.
- Validated contextual variables, such as environmental measurements, only where coverage and meaning support their use.

No feature may use information that would not be known at forecast issue time. Future target values, aggregates containing future observations, and transformations fitted using validation/test periods are prohibited. The target must not accidentally be included as a contemporaneous input for the period being predicted.

5. Data treatment and exclusions

- Keep the frozen dataset unchanged and record its version/checksum in modelling outputs.
- Create target-specific hourly datasets separately, retain source timestamps, zone/series identifiers, quality flags and energy_source where relevant.
- Do not difference cumulative counters across the 2023–2025 gap. The energy target is energy_Wh, not a difference calculated across reset-prone cumulative counters Ea/Er.
- Do not automatically interpret missing values as zero, and do not create unsupported observations to fill gaps.
- Treat the source periods as distinct regimes during profiling and evaluation. Any combined-regime model must be justified by validation evidence.
- Exclude or separately analyse series with inadequate coverage, ambiguous measurement semantics, or target values that cannot be constructed reliably; document the reason.
- Do not claim money saved, carbon avoided, or confirmed waste unless tariff/emissions information and validation evidence are available.

6. Model strategy to be tested

The modelling team should not decide in advance that one global model or one model per zone is always best. For each target, compare feasible alternatives using the same chronological evaluation design:

1. A simple baseline (for example, the value from the same hour on the previous week, or a weekday/hour historical median where enough data exists).
2. At least one suitable machine-learning candidate, selected according to data volume, target behaviour and implementation constraints.
3. A pooled/global model across eligible series versus per-zone or per-series models, if sample size supports a fair comparison.

Use MAE as a primary error metric and RMSE as a secondary metric. Report results by target and, where data permits, by zone/series and temporal regime. Avoid relying on MAPE where target values can be zero or close to zero.

7. Decisions reserved for Task 6

Task 6 will define the exact chronological training, validation and test periods, baseline implementation, metric calculation rules, and how the historical and 2025 regimes are evaluated. The test period must remain untouched during model selection and tuning. Exact date boundaries should be chosen after profiling target coverage and hourly completeness; they are not invented in this specification.

8. Risks and open questions

- Check that hourly energy totals are constructed from complete, non-overlapping intervals and that the 2025 minute-level data has adequate coverage.
- Account for the long gap between historical and recent regimes and potential changes in sensors, operating conditions or consumption behaviour.
- Some zones/series have uneven coverage, NE has no environmental-sensor coverage for several variables, so contextual features cannot be assumed available everywhere.
- Three target experiments increase workload. energy_Wh remains mandatory, W is the first secondary target, I is exploratory and may be dropped if its data quality or added value is insufficient.

9. Task 5 Definition of Done

- ☒ The primary target, unit, forecast horizon and hourly resolution are documented.
- ☒ The zone/series scope and eligibility principles are defined.
- ☒ Candidate features, data-treatment rules and leakage exclusions are documented.
- ☒ The distinction between the frozen source dataset and derived modelling datasets is explicit.
- ☒ The open decisions needed for Task 6 are listed and no unsupported split dates or model winner are claimed.

10. Expected deliverable

A shared modelling specification that every team member can use to answer: what is being predicted, why each target matters, at what temporal resolution and horizon, for which eligible zones/series, from which permitted information, and under which data-quality limitations.